

Signal detection and parameter estimation method in the case of noise uncertainty

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Abstract—The signal detection and parameter estimation for communication signals is an important issue in signal processing. Traditional methods lack of sufficient reliability in the condition of significant noise uncertainty. In this paper, we propose a signal detection and parameter estimation method in the case of noise uncertainty. The method first employs spectrum smoothing and difference operations to extract spectral features in the subsequent step. Then the average power of each spectrum segment and the noise power are estimated utilizing a power-based detection judgment criteria. Finally the signal can be detected and the frequency and bandwidth can be estimated accurately. Numerical results demonstrate the effectiveness of the proposed method.

Index Terms—signal detection, parameter estimation, noise uncertainty, spectrum smoothing, detection probability

I. INTRODUCTION

Signal detection and parameter estimation are crucial due to growing demand for limited spectrum [1], [2]. Signal detection is the basis step of the signal analysis [3], [4]. Various traditional schemes have been proposed for signal detection, such as matched filter (coherent, requiring prior knowledge [5]), cyclostationary feature detection (exploiting signal periodicity [6], [7]), and low-complexity energy detection [8]–[10]. However, the performance of these methods degrade under noise uncertainty [11] due to the noise fluctuation. Subsequently, variety of noise power estimation-based methods have been introduced. Existing techniques include analyzing noise uncertainty impact analyzing method [12], noise variance estimation-based thresholding method [10], and double-threshold methods [13]–[15] are successively pro-

posed. However, the detection performance of these methods degrade greatly in the case of significant noise fluctuation.

Recent studies address noise uncertainty with advanced schemes: a two-stage sensing method [16] realizes detection and parameter estimation of BPSK signals under low SNR, a Duffing oscillator-based approach [17] enables ultra-low SNR LFM signal detection with FPGA implementation via frequency period alignment, neural network ensembles [18] enhance communication signal parameter estimation robustness through uncertainty quantification, and an adaptive frequency-domain method [19] optimizes threshold setting for fluctuating noise to improve spectrum sensing performance.

Besides signal detection, fast and accurate communication signal parameter estimation under low SNR is also critical for practical wireless systems [20]–[22]. There are various methods proposed to address this problem in the literature [16], [23]–[26]. A Haar wavelet transform based method was proposed in [23] for estimating the carrier frequency and bandwidth of BPSK signals. A scheme based on cyclic spectrum was proposed to estimate the carrier frequency, bandwidth and symbol rate, which was only conducted in the Gaussian noise environment [24]. A signal detection and parameter estimation method was introduced under low SNR conditions in [26]. However, the above-mentioned methods usually exhibit inferior performance in fluctuating noise scenarios.

In this paper, a signal detection and parameter estimation method is proposed in the case of noise uncertainty. Specifically, the spectrum smoothing and difference operations are performed first in the proposed method. Then, the segment average signal power and judgment criteria is utilized in the method for signal detection and the position of the signal spectrum is obtained via discontinuous point count comparison in the parameter estimation method. Finally, the carrier fre-

This work was supported by the National Natural Science Foundation of China under Grant 62301590, and the Beijing Nova Program (Grant No. 20240484571).

quency and bandwidth of the signal are calculated. Numerical results demonstrate the effectiveness of the proposed methods for signal detection and parameter estimation under noise uncertainty.

The remainder of the paper is organized as follows. The proposed methods are elaborated in Section II. Section III introduces the numerical results. Concluding remarks are given in Section IV.

II. PROPOSED METHOD

A. Signal detection method

Noise fluctuation significantly impacts signal detection performance. To improve SNR, Fast Fourier Transformation (FFT) is applied to the received signal, followed by noncoherent accumulation to widen the signal-noise power gap.

Given fluctuating sampling points in the actual signal spectrum, spectrum smoothing is applied to reduce false alarm probability. The rectangular wave smoothing method adopts an adaptively adjustable sliding window for spectrum convolution, eliminating fluctuations by averaging each sampling point with its neighboring points. Specifically, the smoothed value of the i th sampling point is the average value of $2M+1$ sampling points, that is,

$$y_{smooth}(i) = \frac{1}{2M+1} \sum_{j=-M}^M y(i+j) \quad (1)$$

where $y(i)$ denotes the i th element of the signal vector.

Then, difference operation is applied to smoothed data to extract the spectral features, that is,

$$\tilde{y} = y(D:N) - y(1:N-D+1), \quad (2)$$

where y denotes the signal spectrum, D and N represent a difference operation interval and the spectrum length, respectively.

We assume that the number of spectrum segments is K , and the bandwidth corresponding to each segment is $B_s = F_s/K$, where F_s denoting the sampling rate. The number of sampling points in each spectrum segment is expressed as,

$$N_s = \frac{B_s Q}{F_s}, \quad (3)$$

where Q denotes the number of FFT points.

The average power of each frequency spectrum segment $q(k)$ can be expressed as,

$$q(k) = \frac{\sum \tilde{y}((k-1)N_s + 1 : kN_s)}{\sum (\tilde{y}((k-1)N_s + 1 : kN_s) > 0)}, k = 1, 2, \dots, K. \quad (4)$$

The signal window index i_{max} is determined by searching the maximum average power $q(k)$, $k = 1, 2, \dots, K$.

The noise power can be estimated by calculating the average of the spectral power except the signal window and its adjacent windows, that is,

$$p_n = \begin{cases} \frac{\sum q(1:i_{max}-2) + \sum q(i_{max}+2:K)}{K-3}, & \text{if } K \geq i_{max} + 2, \\ \frac{\sum q(1:i_{max}-2)}{i_{max}-2}, & \text{otherwise.} \end{cases} \quad (5)$$

Signal detection criteria are proposed in the following.

- 1) If $\tilde{y}_{max} < m \cdot p_n$, no signal exists;
- 2) Otherwise, if the window of the second maximum spectrum power k_{sec} is adjacent to the signal window k_{sig} , the signal exists;
- 3) Otherwise, if $\tilde{y}_{max} > n \cdot (q(k_{sig}-1) + q(k_{sig}+1))$, the signal exists.

Note that parameters m and n are set empirically and adaptively for practical scenarios. The smoothing parameter M is selected to balance noise suppression and spectral feature retention. The difference interval D is determined according to the spectral edge variation characteristics of communication signals. The segment number K is reasonably configured to ensure stable statistical calculation of segment power. Implementation details are shown in Algorithm 1.

Algorithm 1: The proposed signal detection algorithm.

Input: The received data y , the sampling rate F_s , the accumulation group number, the length of sliding window $2M+1$, the number of frequency segments K and the difference operation interval D

Output: Signal detection result

- 1 Implement FFT on the received data and obtain the smoothed spectrum per Eq. (1);
 - 2 Obtain the difference spectrum per Eq. (2);
 - 3 Set the count number c and d to be zero;
 - 4 **for** $h = 1 : 2$ **do**
 - 5 **for** $k = 1 : K$ **do**
 - 6 Calculate the average power of each frequency spectrum segment $q(k)$ per Eq. (4);
 - 7 Estimate the noise power p_n per Eq. (5);
 - 8 If $\tilde{y}_{max} < m \cdot p_n$, $c = c + 1$; otherwise, if $k_{sec} = k_{sig} - 1$ or $k_{sec} = k_{sig} + 1$, $d = d + 1$; otherwise, if $\tilde{y}_{max} > n \cdot (q(k_{sig}-1) + q(k_{sig}+1))$, $d = d + 1$; otherwise, $c = c + 1$;
 - 9 **end**
 - 10 If $d > 0$, the signal exists; otherwise, signal does not exist;
 - 11 **end**
 - 12 Return signal detection result.
-

B. Parameter estimation method

In the following, carrier frequency and bandwidth are estimated based on the detected signal. First, non-coherent accumulation and difference operation are performed as in the signal detection method. Subsequently, the signal spectrum length is coarsely estimated by comparing spectrum power with the threshold. The threshold thr_{smo} is defined as $thr_{smo} = \tau_1 \tilde{y}_{max}$ with τ_1 being predefined. Meanwhile, the discontinuous interval d_{in} which is the consecutive points that do not exceed the threshold is predefined empirically. Then, we introduce the following judgment criteria. If the

current frequency power is greater than the smoothing threshold thr_{smo} , the signal spectrum is continuous. Otherwise, there are two situations where the current frequency power is less than the smoothing threshold thr_{smo} . In one case, the current frequency point is the termination position of the signal spectrum. The other case is that the power of the current frequency point does not exceed the smoothing threshold thr_{smo} due to the frequency fluctuations. Therefore, in order to avoid misjudgment caused by the second situation, the position is determined as the end position of the signal spectrum when the number of consecutive points is less than the predefined discontinuous interval d_{in} .

Finally, the frequency corresponding to the middle point of the start and end position of the spectrum is regarded as the carrier frequency, and the spectrum corresponding to the start and end position of the spectrum is regarded as the bandwidth. The implementation detail is shown in Algorithm 2.

Algorithm 2: The proposed parameter estimation algorithm.

Input: The received data y , the sampling rate F_s , the accumulation group number, the length of sliding window $2M + 1$, the number of frequency segments K , the difference operation interval D and the discontinuous interval d_{in}

Output: Estimated carrier frequency and bandwidth

- 1 Implement FFT on the received data and obtain the smoothed spectrum per Eq. (1);
 - 2 Obtain the difference spectrum per Eq. (2);
 - 3 Set the smoothing threshold $thr_{smo} = \tau_1 \tilde{y}_{max}$;
 - 4 Set the difference threshold $thr_{dif} = \tau_2 \tilde{y}_{max}$;
 - 5 $\bar{y}_1(\tilde{y} \geq thr_{dif}) = 1$, $\bar{y}_2(\tilde{y} \leq -thr_{dif}) = 1$ and $\bar{y} = \bar{y}_1 + \bar{y}_2$;
 - 6 Set the count number c and d to be zero; Set the discontinuous frequency point number g to be zero;
 - 7 **for** $t = 1 : N$ **do**
 - 8 If $\tilde{y}(t) \geq thr_{smo}$, $idx(t) = 1$; otherwise, $idx(t) = 0$; If $d = 0$ and $idx(t) = 1$, $d = d + 1$;
 - 9 otherwise, if $idx(t) = 1$, $d = d + 1$;
 - 10 otherwise, if $g < d_{in}$, $d = d + 1$, $g = g + 1$;
 - 11 otherwise, $f(:, 1) = d - g - 1$, $f(:, 3) = t - g - 1$,
 $f(:, 2) = f(:, 3) - f(:, 1)$,
 $f(:, 4) = \sum(\bar{y}(f(:, 2) : f(:, 3)))$;
 - 12 **end**
 - 13 Set the frequency $freq = (-N/2 : N/2 - 1)/N \cdot F_s$;
 - 14 Count the number $J(f(:, 1) \neq 0)$; Sort $f(:, 4)$ with descending order, record the values and corresponding indices as u and v , respectively;
 - 15 **for** $j = 1 : J$ **do**
 - 16 $f_{start} = f(v(j), 2)$; $f_{stop} = f(v(j), 3)$;
 $f(j) = freq((f_{start} + f_{stop})/2)$;
 $bw(j) = freq(f_{stop}) - freq(f_{start})$;
 - 17 **end**
 - 18 Return the estimated carrier frequency f and bandwidth bw .
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III. NUMERICAL RESULTS

A. Simulation results

To examine the impact of accumulation group number on signal detection, the detection probability versus SNR in the case of different group numbers is shown in Fig. 1. Simulation parameters are set as follows. The bit rate of the BPSK signal is set as 3.25MHz. The shaping filter factor is set as 0.3. The sampling rate is 32.5MHz. The sliding window length and the frequency segment number are set as 16 and 8, respectively. The difference operation interval is defined as 16. 200 Monte-Carlo trials are utilized to calculate the detection probability. When $SNR = -10$ dB, detection probability reaches 95% when the group number exceeds 10. Sufficient spectrum accumulation can effectively compensate for signal energy attenuation under low-SNR conditions.

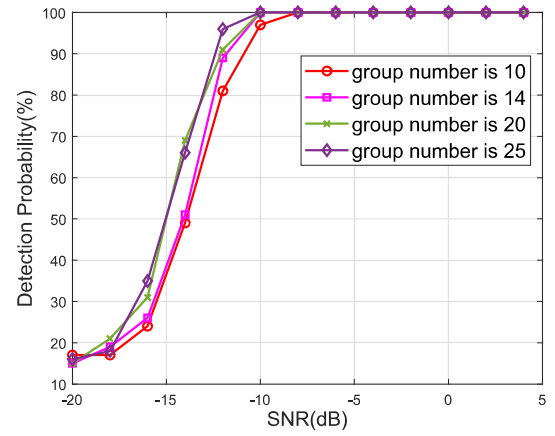


Fig. 1. Detection probability of the proposed method versus SNR in the case of different accumulation group number.

B. Experimental results

The numerical experimental results are given in the following. The physical equipment diagram is shown in Fig. 2.



Fig. 2. Physical equipment diagram.

In the first example, a BPSK signal (carrier frequency 5.85GHz) is down-converted to 550MHz and the output frequency is -2.5MHz. Signal source output power is -67dBm. The attenuation is set as 39.04dBm and the actual received power is -106.04dBm. The bit rate is 1Mbps and the sampling rate is 61.44MHz. In the experiments, 14 accumulation group and 8 frequency segments are utilized. The sliding window

length is set as 16. The smoothed frequency spectrum is shown in Fig. 3(a). The difference operation interval is set to 22 and the frequency spectrum after difference operation is shown in Fig. 3(b). The estimated carrier frequency is -2.49MHz and the frequency bandwidth is 0.78MHz, which are approach to the true values.

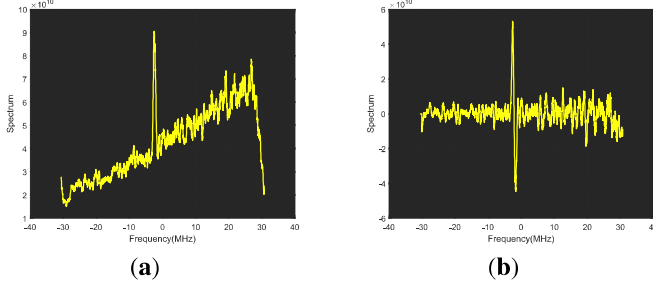


Fig. 3. Spectrum of proposed method in first example: (a) smoothed spectrum. (b) difference spectrum.

In the second example, a BPSK signal (carrier frequency 5.86GHz) is down-converted to 560MHz and the output frequency is 7.5MHz. The signal source output power is -66dBm. The attenuation is set as 39.04dBm and the actual received power is -105.04dBm. The bit rate is 1.5Mbps. The sampling rate is defined same with the first example. The smoothed spectrum is shown in Fig. 4(a), and the difference spectrum in Fig. 4(b). The carrier frequency is estimated as 7.53MHz and bandwidth is 1.65MHz.

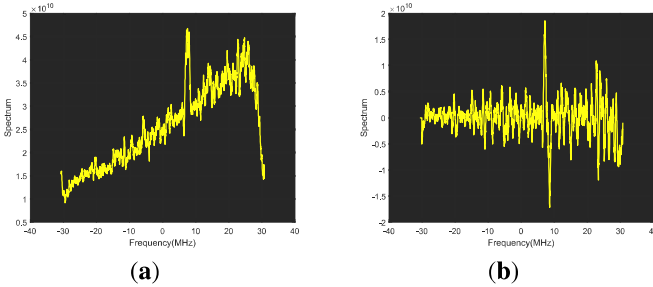


Fig. 4. Spectrum of proposed method in second example: (a) smoothed spectrum. (b) difference spectrum.

In the third example, a BPSK signal (carrier frequency 5.87GHz) is down-converted to 570MHz and the output frequency is 17.5MHz. The signal source output power is -61dBm. The attenuation is set as 39.04dBm and the actual received power is -100.04dBm. The bit rate is 4Mbps. Other parameters is set as the same with the first example. The smoothed spectrum is shown in Fig. 5(a), and the difference spectrum is given in Fig. 5(b). The carrier frequency is estimated as 17.64MHz and the bandwidth is 4.35MHz, which verify the effectiveness of the proposed method.

Various results are given in Table I and Table II. A total of 100 experiments are conducted. When the received power is -105.04 dBm, intermediate frequency is -2.5 MHz and bit rate is 1 Mbps, the RMSE of carrier frequency is 0.08 MHz. According to the experiments, the signals can be detected and

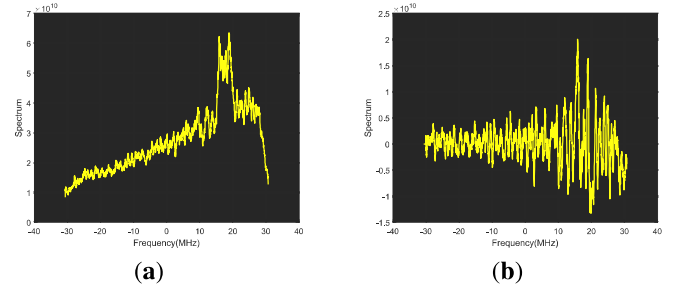


Fig. 5. Spectrum of proposed method in third example: (a) smoothed spectrum. (b) difference spectrum.

parameters can be estimated accurately, which validate the effectiveness of the proposed method.

TABLE I
SIGNAL DETECTION

Radio frequency (GHz)	Output power (dBm)	Receive Power (dBm)	Intermediate frequency (MHz)	Bit rate (Mbps)	Signal detection results
5.85	-66	-105.04	-2.5	1	✓
5.85	-67	-106.04	-2.5	1	✓
5.85	-68	-107.04	-2.5	1	×
5.86	-65	-104.04	7.5	1.5	✓
5.86	-66	-105.04	7.5	1.5	✓
5.86	-67	-106.04	7.5	1.5	×
5.87	-61	-100.04	17.5	4	✓
5.87	-62	-101.04	17.5	4	×

✓ and × indicate that a signal is detected or not detected, respectively.

TABLE II
PARAMETER ESTIMATION

Receive Power (dBm)	Intermediate frequency (MHz)	Bit rate (Mbps)	Estimated results	
			frequency (MHz)	bandwidth (MHz)
-105.04	-2.5	1	-2.55	0.93
-106.04	-2.5	1	-2.49	0.78
-107.04	-2.5	1	\	\
-104.04	7.5	1.5	7.44	1.83
-105.04	7.5	1.5	7.53	1.65
-106.04	7.5	1.5	\	\
-100.04	17.5	4	17.64	4.35
-101.04	17.5	4	\	\

\ indicates the correct results cannot be estimated.

IV. CONCLUSION

A signal detection and parameter estimation method is proposed to improve the performance in the case of noise uncertainty. Traditional methods suffer significant degradation due to random noise fluctuation. In order to overcome the drawback, the spectrum smoothing and difference operations were introduced to mitigate noise interference fundamentally. Two different types of judgment criteria were developed for signal detection and parameter estimation accurately. Numerical results show high detection and estimation accuracy, verifying effectiveness of the proposed method in the condition of noise uncertainty.

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